Semester 1, 2026

Business Plan and Engineering Design Progress Report

EGH419: Advanced Design and Entrepreneurship

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1 Product Name

The proposed product developed by Group 40 is titled **PowerStake**. PowerStake is a smart power management system designed to help households reduce electricity costs through appliance-level monitoring, automated outlet control, predictive insights, and a user-friendly mobile dashboard. The concept combines affordable hardware with intelligent software to reduce standby energy waste, improve convenience, and identify potential appliance faults before failure occurs.

The Group 40 project team is structured using functional roles to allocate responsibilities across business, technical, and design tasks. Team members and primary subsystem responsibilities are listed below:

- **Handler - Erin Keur:** Project coordination and power usage reduction strategy.
- **Hustler - Ethan Bailey:** Commercial strategy, customer value proposition, and mobile dashboard development.
- **Hacker - Andrew Pollard:** Hardware development including power measurement and supply systems.
- **Hipster - Xanthea Vazey:** User experience, industrial design, and automatic power-off scheduling functions.
- **Helper - Jeremy Khin Zaw:** Cross-functional support across all project areas, with primary responsibility for fault detection systems.

This team structure enables parallel development of the core PowerStake subsystems while maintaining shared ownership of prototype integration and validation activities.

2 Target TRL

The current readiness level of the Automatic Power-off Scheduling subsystem is estimated at TRL 2. At present, the concept has been defined and supported through preliminary research into existing technologies such as smart plugs, programmable timers, relay modules, and mobile-controlled outlet systems. However, no integrated prototype has yet been constructed or experimentally validated by the project team.

Given the remaining project schedule, the realistic target for this assessment is TRL 3 progressing toward early TRL 4. This would involve developing and demonstrating a proof-of-concept prototype using commercially available microcontrollers (e.g. ESP32), relay modules, and basic power sensing inputs. Initial validation may be performed on low-voltage test circuits or simulated appliance loads rather than direct 240 V mains systems, reducing safety risk and development complexity.

A successful prototype outcome would demonstrate the core subsystem functions of detecting inactive loads, applying simple shutoff logic, manually restoring power, and communicating with a mobile dashboard interface. Supporting work may also include enclosure concepts developed through CAD modelling and early physical packaging studies for future iterations.

Achieving a full TRL 4 mains-ready prototype within the current timeframe is considered ambitious due to hardware procurement, electrical safety constraints, and software integration requirements. A more practical near-term pathway would be completion of a validated bench-scale prototype during this unit, followed by future progression to TRL 4-5 through controlled

household testing across common appliances such as televisions, chargers, gaming consoles, and kitchen devices.

3 Problem Validation and Engineering Design

3.1 Customer Needs

Many households waste electricity through standby loads from televisions, chargers, gaming consoles, printers and kitchen appliances [1]. While individual standby loads are small, cumulative yearly costs are meaningful. Existing timer switches require manual programming and do not adapt to user behaviour. Most smart plugs offer remote switching but still depend on active user involvement. Customers require a solution that automatically reduces unnecessary consumption while maintaining convenience. The system must be simple to use, safe, and allow manual override when required.

3.2 High-Level Objective (HLO-4)

The subsystem shall automatically recommend or execute outlet shutoff schedules that reduce standby energy consumption by at least 40% during controlled testing, while allowing manual user override within 2 seconds.

4 System Requirements

4.1 Physical Requirements

ID	**Requirement**	**HLO**
PH-01	Subsystem shall control a minimum of four independent outlets.	HLO-4
PH-02	Subsystem shall operate from 240V Australian mains supply via an isolated relay module.	HLO-4
PH-03	Subsystem shall include a Wi-Fi capable microcontroller for communication and control.	HLO-4
PH-04	Prototype enclosure footprint shall be less than 350 mm $\times$ 150 mm.	HLO-4

Table 1: Physical system requirements

4.2 Functional Requirements

ID	Requirement	HLO
FN-01	Subsystem shall measure outlet power state once per second.	HLO-4
FN-02	Subsystem shall detect standby condition below a configurable wattage threshold.	HLO-4
FN-03	Subsystem shall generate shutoff schedules using repeated inactivity patterns.	HLO-4
FN-04	Subsystem shall allow manual override through the mobile dashboard.	HLO-4
FN-05	Subsystem shall restore power immediately upon authorised user command.	HLO-4

Table 2: Functional system requirements

4.3 Performance Requirements

ID	Requirement	HLO
PE-01	Subsystem shall reduce standby energy consumption by at least 40% during controlled testing.	HLO-4
PE-02	Switching response time shall be less than 2 seconds after command input.	HLO-4
PE-03	Schedule recommendations shall achieve at least 80% user acceptance accuracy.	HLO-4
PE-04	Subsystem shall operate continuously for 30 days without requiring reset.	HLO-4

Table 3: Performance system requirements

4.4 Subsystem Architecture

The Automatic Power-off Scheduling subsystem forms part of the wider PowerStake platform by linking power monitoring, intelligent control, wireless communication, and user interaction. Power sensors provide real-time outlet usage data which is processed by the onboard microcontroller. A practical prototype could utilise an ESP32 microcontroller due to its integrated Wi-Fi capability, sufficient processing performance, and low-cost implementation [2]. As well as ACS712 current sensing modules, and isolated relay switching hardware to provide low-cost monitoring and automated outlet control [3].

The controller stores recent usage history and applies scheduling logic to identify repeated inactivity periods or standby behaviour. When suitable conditions are met, relay drivers disconnect power to selected outlets in order to reduce unnecessary energy consumption.

The Wi-Fi module enables two-way communication between the device and the mobile dashboard. This allows users to manually override schedules, restore outlet power, and review system recommendations. In later development stages, cloud-based analysis could be introduced to improve shutoff recommendations using longer-term usage trends.

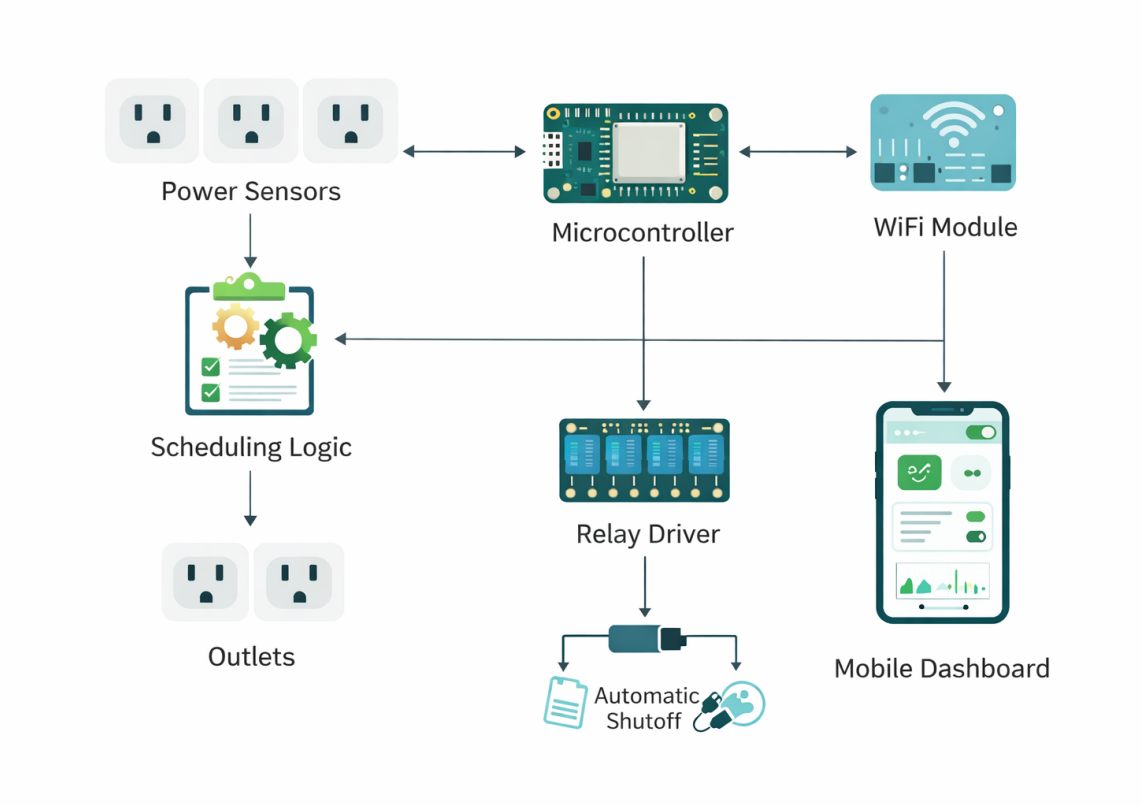

Figure 1: System Architecture of the Automatic Power-off Scheduling Subsystem [4].

This subsystem interfaces directly with the wider PowerStake monitoring, control, and mobile dashboard environment.

4.5 Trade Studies

A trade study was conducted to determine the most suitable switching method for the Automatic Power-off Scheduling subsystem. Reliable switching is essential, as the selected component must disconnect outlet power when standby conditions or scheduled shutoff periods are detected. For the current prototype stage, the preferred solution needed to be low-cost, safe, easy to integrate with an ESP32 microcontroller, and practical to implement within the available project timeframe.

Three switching approaches were evaluated: (1) electromechanical relay module, (2) solid state relay (SSR), and (3) modified commercial smart plug hardware. These options were compared based on switching capability, cost, integration complexity, prototype suitability, and overall practicality.

Option	Switching Capability	Cost (AUD)	Integration Complexity	Suitability
Mechanical Relay Module	10A, 250VAC switching contacts [5]	$8–15	Low – direct GPIO control, common modules available	High
Solid State Relay (SSR)	Silent AC switching, high cycle life [6]	$15–30	Medium – load matching and thermal management required	Medium
Modified Smart Plug Hardware	Existing mains switching in consumer enclosure [7]	$20+	High – proprietary PCB, difficult modification	Low

Table 4: Automatic Power-off Scheduling trade study

The mechanical relay module was selected as the preferred solution due to its low cost, simple control interface, and suitability for rapid prototype development. Standard relay modules can be controlled directly using ESP32 digital outputs and provide galvanic isolation between the low-voltage control circuit and the switched load [5]. This makes them practical for early-stage testing and subsystem validation.

Solid state relays provide silent operation and long switching life; however, they are more expensive and may require additional thermal management depending on the connected load [6]. While technically strong, they were considered less practical for an initial university prototype.

Modified smart plug hardware was the least suitable option. Although it contains integrated switching hardware, internal circuitry is often proprietary and not intended for modification, increasing development time and reducing transparency during testing [7].

Overall, the mechanical relay module offered the best balance of cost, safety, ease of integration, and functional performance for achieving the subsystem High-Level Objective within the current project scope.

References

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